

Subhav Kumar

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EDUCATION

B.Tech in Computer Science & Engineering
Indian Institute of Information Technology (IIIT), Raichur

2023 – 2027
CGPA: 8.99 / 10.00

EXPERIENCE

AI Research Intern — WSAI, IIT Madras May 2026 – Present

- Researching **Semantic Text Reconstruction from Embeddings** — investigating decoder-based methods to recover meaningful text from dense vector representations.
- Building evaluation tooling in **Python** to measure reconstruction fidelity, semantic similarity, and latent space geometry across transformer architectures.
- Running large-scale experiments on **HPC/GPU clusters**, optimizing training pipelines for throughput and memory efficiency.

RESEARCH

Paper Accepted — IEEE ICCSE 2026 Budapest, Hungary

- *"PCA-based Defense Against Embedding Inversion and User History Leakage in Recommendation Systems."*
- Proposed a **PCA-based dimensionality reduction** defense that mitigates embedding inversion and membership inference attacks on recommendation system user embeddings.
- Preserved recommendation quality while significantly reducing attacker reconstruction accuracy.

PROJECTS

Semantic Text Reconstruction from Embeddings Ongoing

- Built a decoder-only transformer pipeline in **PyTorch** to reconstruct natural language text from sentence embeddings.
- Implemented custom training loops with attention masking, embedding injection, and beam search decoding.
- Evaluated output using BLEU, cosine similarity, and semantic equivalence metrics across multiple embedding models.
-  [View Project Repository](#)

Student Analytics Dashboard Aug – Sep 2025

- Developed a full-stack analytics platform using **React**, **Node.js**, **Express**, and **MongoDB** with interactive data visualizations.
- Implemented **JWT authentication** and **Role-Based Access Control (RBAC)** for secure multi-user access.
- Built RESTful APIs with input validation, error handling, and optimized database queries.
-  [View Project Repository](#)

CCTV Face Enhancement System Nov 2025 – Jan 2026

- Engineered a **computer vision** pipeline to detect, crop, and enhance low-resolution faces from CCTV footage using super-resolution models.
- Integrated face detection (**OpenCV**) with upscaling modules; optimized inference latency for near real-time processing.
-  [View Project Repository](#)

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C/C++, SQL
AI / ML	PyTorch, TensorFlow, Scikit-learn, Transformers, NLP, Computer Vision, LLM Fine-tuning (LoRA)
Web Dev	React, Next.js, Node.js, Express, FastAPI, Django, REST APIs, HTML/CSS, TailwindCSS
Tools & Infra	Git, Docker, Linux, AWS, MongoDB, PostgreSQL, HPC/GPU Clusters

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate
Master SQL and DBMS — Scaler

 [Verify](#)
 [Verify](#)

ACHIEVEMENTS

- **Amazon ML Challenge 2025:** Ranked in **Top 1200** out of 80,000+ participants.
- **Competitive Programming:** Solved **400+ DSA problems** (LeetCode/GFG/Codeforces). 
- **Hackathons:** Walmart Sparkathon  and Adobe India Hackathon. 
- **PRMO Qualified 2019.**